



HUMAN EYE -03

Class 10 - Science

Section A

1. **State True or False:** [19]
- (a) The prism splits the incident white light into a band of colours called spectrum. [1]
 - (b) The retina is a delicate membrane with having an enormous number of light-sensitive cells. [1]
 - (c) The band of seven colours formed on a white screen, when a beam of light is passed through a glass prism, is called spectrum of white light. [1]
 - (d) A rainbow is a natural spectrum appearing in the sky after a rain shower. [1]
 - (e) Myopia or near-sightedness is an eye defect in which a person can see nearby objects clearly but cannot see distant objects distinctly. [1]
 - (f) The minimum distance, at which objects can be seen most distinctly without strain, is called the least distance of distinct vision or the near point of the eye. [1]
 - (g) When a beam of light strikes fine particles of matter, the path of the beam becomes visible. [1]
 - (h) The scattering of light by particles in its path is called the Tyndall effect. [1]
 - (i) If the size of the scattering particles is large enough, then, the scattered light may even appear black. [1]
 - (j) Retina is present behind the cornea. [1]
 - (k) For a young adult with normal vision, the near point is about 25 cm. [1]
 - (l) The lens of the human eye forms an image on a light-sensitive screen called the pupil. [1]
 - (m) The refraction of light caused by the earth's atmosphere is called atmospheric refraction. [1]
 - (n) The ability of the eye lens to adjust its focal length is called accommodation. [1]
 - (o) Myopia is a condition in which the crystalline lens of people at old age becomes milky and cloudy resulting in blurred vision. [1]
 - (p) Presbyopia is a human eye defect because of which an old person cannot read and write comfortably. [1]
 - (q) Iris is a dark muscular diaphragm that controls the size of the pupil. [1]
 - (r) The splitting up of white light into seven colours on passing through a transparent medium like a glass prism is called a spectrum of light. [1]
 - (s) Light enters the eye through a thin membrane called the cornea. [1]

Section B

2. **Fill in the blanks:** [19]
- (a) _____ of light means to throw light in various random directions. [1]
 - (b) _____ is the phenomenon of bending of light on passing through the Earth's atmosphere. [1]
 - (c) The white light is dispersed into _____ colour components by a prism. [1]
 - (d) The ability of the eye lens to adjust its focal length is called _____. [1]
 - (e) _____ is an eye defect in which a person can see nearby objects clearly but cannot see distant [1]

objects distinctly.

- (f) The scattering of light by particles in its path is called _____. [1]
- (g) The angle between the incident ray and emergent ray is called the angle of _____. [1]
- (h) _____ is a human eye defect because of which an old person cannot read and write comfortably. [1]
- (i) The splitting up of white light into seven colours on passing through a transparent medium like a glass prism is called _____ of light. [1]
- (j) _____ is present behind the cornea. [1]
- (k) _____ is a condition in which the crystalline lens of people at old age becomes milky and cloudy resulting in blurred vision. [1]
- (l) _____ is formed by the dispersion of sunlight by tiny water droplets, present in the atmosphere. [1]
- (m) The prism splits the incident white light into a band of colours is called its _____. [1]
- (n) The _____ is a delicate membrane having an enormous number of light-sensitive cells. [1]
- (o) The twinkling of a star is due to atmospheric _____ of starlight. [1]
- (p) _____ is a hole in the middle of the iris which adjusts its opening to allow the required amount of light for clearing purpose. [1]
- (q) The time difference between the actual sunset and the apparent sunset is about _____. [1]
- (r) _____ was the first to use a glass prism to obtain the spectrum of sunlight. [1]
- (s) To correct myopia, the person has to wear spectacles with a _____ lens of focal length equal to the distance of a far point of the myopic eye. [1]

Section C

- 3.
 - i. List three common refractive defects of vision. Suggest the way of correcting these defects. [5]
 - ii. About 45 lac people in the developing countries are suffering from corneal blindness. About 30 lac children below the age of 12 yr suffering from this defect can be cured by replacing the defective cornea with the cornea of a donated eye. How and why can students of your age involve themselves to create awareness about this fact among people?
- 4. How do eye focus objects at different distances ? [5]
- 5. What is **dispersion of white light**? Draw a labelled diagram to illustrate the recombination of the spectrum of white light. Why is it essential that the two prisms used for the purpose should be identical and placed in an inverted position with respect to each other? [5]
- 6. The near point of a hypermetropic eye is 1 m. What is the power of the lens required to correct this defect? Assume that the near point of the normal eye is 25 cm. [5]
- 7. What is dispersion of light? What is its cause? [5]
- 8. Make a diagram to show how hypermetropia is corrected. The near point of a hypermetropic eye is 1 m. What is the power of the lens required to correct this defect? Assume that the near point of the normal eye is 25 cm. [5]
- 9. What is prism? What is its action? Write down the relation between angle of incidence, angle of emergence, angle of prism and angle of deviation. [5]
- 10. When do we consider a person to be myopic or hypermetropia? List two causes of hypermetropia. Explain using ray diagrams how the defect associated with hypermetropia eye can be corrected. [5]
- 11. What is meant by scattering of light? Mention the factor on which it depends. Explain, why the colour of the clear sky is blue? An astronaut in space finds sky to be dark. Explain reason for this observation. [5]
- 12. A person is not able to see far off objects clearly. From which defect of the eye the person is suffering ? How to correct this defect ? [5]

13. Differentiate between myopia and hypermetropia. [5]
14. a. Write the function of each of the following parts of human eye: [5]
 - i. Cornea
 - ii. Iris
 - iii. Crystalline lens
 - iv. Ciliary muscles

b. Why does the sun appear reddish early in the morning? Will this phenomenon be observed by an astronaut on the Moon? Give reason to justify your answer.
15. What are rods and cones ? Give their action. [5]
16. With the help of well labeled diagram, explain the construction and working of human eye. [5]
17. Describe Newton's disc experiment to show that white light is composed of seven spectrum colours. [5]

Why seven colours combine to give almost white but not perfectly white ?
18. How can we explain the reddish appearance of the sun at sunrise or sunset? Why does it not appear red at noon? [5]
19. a. What is myopia? State the two causes of myopia. With the help of labelled ray diagrams show [5]
 - i. the eye defect myopia
 - ii. correction of myopia using a lens.

b. Why is the normal eye unable to focus on an object placed within 10 cm from the eye?
20. Differentiate between dispersion and scattering of light. [5]
21. What is atmospheric refraction? Use this phenomenon to explain the following natural events. [5]
 - i. Twinkling of stars.
 - ii. Advanced sunrise and delayed sunset.

Draw diagrams to illustrate your answers.
22. A 14-year old student is not able to see clearly the questions written on the blackboard placed at a distance of 5 [5]
m from him.
 - a. Name the defect of vision he is suffering from.
 - b. With the help of labelled ray, diagrams show how this defect can be corrected.
 - c. Name the type of lens used to correct this defect.
23. Explain the refraction of light through a triangular glass prism using a labelled ray diagram. Hence define the [5]
angle of deviation.
24. a. What is presbyopia? State its cause. How is it corrected? [5]

b. Why does the Sun appear reddish early in the morning? Explain with the help of a labelled diagram.
25. How does refraction take place in the atmosphere? Why do stars twinkle but not the planets? [5]
26. A 14 year old student is not able to see clearly the questions written of the black board placed at a distance of 5 [5]
m from him.
 - a. Name the defect of vision he is suffering from?
 - b. Draw the diagram to show this defect?
 - c. Name the type of lens used to correct this defect?
 - d. Name two possible cause of this defect.
 - e. Draw the diagram to show how this defect can be corrected.
27. What is meant by scattering of light? Explain Tyndall effect. Give a few illustrations of scattering of light. [5]
28. Due to gradual weakening of ciliary muscles and diminishing flexibility of the eye lens, a certain defect of vision [5]

arises. Write the name of this defect. Name the type of lens required by such persons to improve the vision. Explain the structure and function of such a lens.

29. When do we consider a person to be myopic or hypermetropic? Explain using diagrams how the defects associated with myopic and hypermetropic eye can be corrected? [5]
30. A person needs a lens of power -5.5 diopters for correcting his distant vision. For correcting his near vision he needs a lens of power +1.5 diopter. What is the focal length of the lens required for correcting (i) distant vision, and (ii) near vision? [5]
31. a. A student is unable to see clearly the words written on the black board placed at a distance of approximately 3 m from him. Name the defect of vision the boy is suffering from. State the possible causes of this defect and explain the method of correcting it. [5]
b. Why do stars twinkle? Explain.
32. A person is unable to read a book properly. From which defect is he suffering? How to correct this defect? [5]
33. How does atmospheric refraction affect sunrise and sunset? [5]
34. What is hypermetropia? Write two causes for development of this defect Describe with a ray diagram how this defect of vision can be corrected by using spectacles. [5]
35. i. Demonstrate an activity with a well labelled diagram to prove that white light is made up of seven colours. [5]
ii. Which colour of light bends least and which one the most while passing out from the prism? Also, state the reason for the same.
36. Explain the structure and functioning of the human eye. How are we able to see nearby as well as distant objects? [5]
37. Differentiate between presbyopia and cataract. [5]